

D-SCRAMBLED CANTOR SETS

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ABSTRACT. For all $d \geq 3$, we provide an example of a homeomorphism of Cantor space that has an uncountable d -scrambled set but no d -scrambled Cantor set.

INTRODUCTION

Suppose that X is a metric space and $f: X \rightarrow X$. We say that $x \in X^d$ is *proximal* if $\liminf_{n \rightarrow \infty} \max_{i < j < d} d_X(f^n(x(i)), f^n(x(j))) = 0$, *asymptotic* if $\limsup_{n \rightarrow \infty} \min_{i < j < d} d_X(f^n(x(i)), f^n(x(j))) = 0$, and *Li–Yorke* if it is proximal but not asymptotic. A set $Y \subseteq X$ is *d -scrambled* if every injective sequence in Y^d is Li–Yorke, and f is *Li–Yorke d -chaotic* if there is an uncountable d -scrambled set $Y \subseteq X$.

In [GGM21], it was shown that every Li–Yorke 2-chaotic continuous function on a Polish space has a 2-scrambled Cantor set. Here we show that the analogous statement for $d \geq 3$ is false:

Theorem 1 (ZFC). *Suppose that $d \geq 3$. Then there is a Li–Yorke d -chaotic homeomorphism of $2^{\mathbb{N}}$ with no d -scrambled Cantor set.*

A *graph* on X is an irreflexive symmetric set $G \subseteq X \times X$. Let $\text{Hyp}^d(G)$ be the d -ary relation on X of all injective sequences $x \in X^d$ such that $G \cap (x[d] \times x[d]) \neq \emptyset$. An *embedding* of a d -ary relation R on X into a d -ary relation S on Y is an injection $\pi: X \rightarrow Y$ such that $x \in R \iff \pi \circ x \in S$ for all $x \in X^d$. The *saturation* of a set $Y \subseteq X$ under a bijection $T: X \rightarrow X$ is given by $[Y]_T = \bigcup_{n \in \mathbb{Z}} T^n[Y]$. A subset of a topological space is F_σ if it is a countable union of closed sets. The primary new technical result underlying our proof of Theorem 1 is:

Theorem 2 (ZF + DC). *Suppose that $d \geq 3$ and G is an F_σ graph on $2^{\mathbb{N}}$. Then there is a continuous embedding $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ of $\text{Hyp}^d(G)$ into the d -ary Li–Yorkeness relation of a homeomorphism $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ for which the saturation of $\pi[2^{\mathbb{N}}]$ is co-countable and dense.*

In §1, we establish Theorem 2. In §2, we combine this with known facts to establish Theorem 1 and a related independence phenomenon.

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1. EMBEDDING HYPERGRAPHS

For all $s \in 2^{<\mathbb{N}}$, define $\mathcal{N}_s = \{c \in 2^{\mathbb{N}} \mid \forall i < |s| \ s(i) = c(i)\}$. For each metric space X , define $T_X: X^{\mathbb{Z}} \rightarrow X^{\mathbb{Z}}$ by $T_X(x)(i) = x(i+1)$ for all $i \in \mathbb{Z}$ and $x \in X$. As Brouwer's theorem ensures that every non-empty compact perfect zero-dimensional metric space is homeomorphic to $2^{\mathbb{N}}$ (see, for example, [Kec95, Proposition 7.4]), Theorem 2 is a consequence of the following fact:

Theorem 1.1 (ZF + DC). *Suppose that $d \geq 3$ and G is an F_σ graph on $2^{\mathbb{N}}$. Then there exist a compact countable ultrametric space X , a T_X -invariant closed set $F \subseteq X^{\mathbb{Z}}$ whose finite sequences are proximal, and a continuous embedding $\pi: 2^{\mathbb{N}} \rightarrow X^{\mathbb{Z}}$ of $\text{Hyp}^d(G)$ into the complement of the d -ary asymptoticity relation for which the saturation of $\pi[2^{\mathbb{N}}]$ is a co-countable dense subset of F .*

Proof. Fix closed graphs G_n on $2^{\mathbb{N}}$ for which $G = \bigcup_{n \in \mathbb{N}} G_n$. For each $m \in \mathbb{N}$, let $G_{m,n}$ be the graph on 2^m consisting of all pairs of distinct sequences of the form $(c \upharpoonright m, d \upharpoonright m)$, where $(c, d) \in G_n$, and let $H_{m,n}$ be the $(d-1)$ -ary relation on 2^m of all injective sequences $s \in (2^m)^{d-1}$ such that $s(0) G_{m,n} s(1)$. Fix an infinite set $I \subseteq \mathbb{N}$ such that $|I \cap (I-n)| < \aleph_0$ for all $n > 0$, as well as a bijection $\phi: I \rightarrow \mathbb{N} \cup \bigcup_{m,n \in \mathbb{N}} H_{m,n} \times \{n\}$.

Set $X = \{0, 1\} \cup (d \times \mathbb{N}) \cup \{\infty\}$, define $h_X: X \rightarrow \mathbb{R}$ by setting $h_X(0) = h_X(1) = 2$, $h_X(k, n) = 1/2^n$ for all $k < d$ and $n \in \mathbb{N}$, and $h_X(\infty) = 0$, let ρ_X be the compact ultrametric on X given by $\rho_X(x, y) = \max(h_X(x), h_X(y))$ for all distinct $x, y \in X$, and let ρ be the product metric on $X^{\mathbb{Z}}$ given by $\rho(x, y) = \sum_{i \in \mathbb{Z}} \rho_X(x(i), y(i))/2^{|i|}$.

Define $\pi: 2^{\mathbb{N}} \rightarrow X^{\mathbb{Z}}$ by setting $\pi(c)(i) = \infty$ for all $i \in \mathbb{Z} \setminus I$ and

$$\pi(c)(i) = \begin{cases} c(\phi(i)) & \text{if } \phi(i) \in \mathbb{N}, \\ (k, \phi(i)(1)) & \text{if } k < d-1 \text{ and } c \in \mathcal{N}_{\phi(i)(0)(k)}, \text{ and} \\ (d-1, \phi(i)(1)) & \text{if } c \notin \bigcup_{k < d-1} \mathcal{N}_{\phi(i)(0)(k)} \end{cases}$$

for all $i \in I$, for all $c \in 2^{\mathbb{N}}$. Clearly π is continuous.

Lemma 1.2. *The function π is injective.*

Proof. Suppose that $c, d \in 2^{\mathbb{N}}$ are distinct. Then there exists $n \in \mathbb{N}$ with $c(n) \neq d(n)$. Fix $i \in I$ for which $\phi(i) = n$. Then $\pi(b)(i) = b(n)$ for all $b \in \{c, d\}$, so $\pi(c)(i) \neq \pi(d)(i)$, thus $\pi(c) \neq \pi(d)$. $\square$

A homomorphism from a d -ary relation R on Y to a d -ary relation S on Z is a function $\pi: Y \rightarrow Z$ such that $\pi \circ y \in S$ for all $y \in R$.

Lemma 1.3. *The function π is a homomorphism from $\text{Hyp}^d(G)$ to the complement of the d -ary asymptoticity relation.*

Proof. Suppose that $c \in \text{Hyp}^d(G)$ and fix $j < k < d$ and $n \in \mathbb{N}$ for which $c \in \text{Hyp}^d(G_n)$ and $c(j) G_n c(k)$. For all $m \in \mathbb{N}$, fix $s_m \in (2^m)^{d-1}$ such that $s_m(0) = c(j) \upharpoonright m$, $s_m(1) = c(k) \upharpoonright m$, and $c(\ell) \upharpoonright m \notin s_m[d-1]$ for at most one $\ell < d$. Then $I' = \phi^{-1}(\{(s_m, n) \mid m \in \mathbb{N}\})$ is infinite. But if $i \in I'$, then $\{\pi(c(k))(i) \mid k < d\} = \{(k, n) \mid k < d\}$, so $\inf_{i \in I'} \min_{j < k < d} \rho(T_X^i(\pi(c(j))), T_X^i(\pi(c(k)))) \geq 1/2^n$. $\square$

Lemma 1.4. *The function π is a homomorphism from the complement of $\text{Hyp}^d(G)$ to the d -ary asymptoticity relation.*

Proof. Suppose that $c \in (2^{\mathbb{N}})^d$ and $\pi \circ c$ is not asymptotic. Fix $\epsilon > 0$ such that $I' = \{i \in \mathbb{N} \mid \min_{j < k < d} \rho(T_X^i(\pi(c(j))), T_X^i(\pi(c(k)))) \geq \epsilon\}$ is infinite, and $r \in \mathbb{N}$ with $\sum \{1/2^{|i|} \mid i \in \mathbb{Z} \setminus [-r, r]\} < \epsilon/2$. Our choice of I ensures that $I'' = \{i' \in I' \mid |[i' - r, i' + r] \cap I| \leq 1\}$ is co-finite in I' , so our choice of r ensures that there is a unique $p(i'') \in I \cap [i'' - r, i'' + r]$ for all $i'' \in I''$. Then the set $I''' = p[I'']$ is infinite. Moreover, if $i'' \in I''$, then $\min_{j < k < d} \rho_X(\pi(c(j))(p(i'')), \pi(c(k))(p(i''))) \geq \epsilon/2$. As $d \geq 3$, it follows that $\phi(p(i'')) \notin \mathbb{N}$ and $\sup_{i''' \in I'''} \phi(i''')(1) < \infty$, so there exist an infinite set $I'''' \subseteq I'''$, distinct $j, k < d$, and $n \in \mathbb{N}$ such that $c(j) \in \mathcal{N}_{\phi(i''''(0)(0))}$, $c(k) \in \mathcal{N}_{\phi(i''''(0)(1))}$, and $n = \phi(i''''(1))$ for all $i'''' \in I''''$. As G_n is closed and the lengths of $\phi(i''''(0)(0))$ and $\phi(i''''(0)(1))$ are unbounded, it follows that $c(j) G_n c(k)$, and therefore $c \in \text{Hyp}^d(G_n)$, so $c \in \text{Hyp}^d(G)$. $\square$

Let F be the closure of the T_X -saturation of $\pi[2^{\mathbb{N}}]$.

Lemma 1.5. *Suppose that $x \in F \setminus [\pi[2^{\mathbb{N}}]]_{T_X}$. Then $|\mathbb{Z} \setminus x^{-1}(\{\infty\})| \leq 1$.*

Proof. We will show that $|[-i, i] \setminus x^{-1}(\{\infty\})| \leq 1$ for all $i \in \mathbb{N}$. We can assume that there exist a sequence $(c_n)_{n \in \mathbb{N}}$ of elements of $2^{\mathbb{N}}$, $\epsilon \in \{\pm 1\}$, and a strictly increasing sequence $(i_n)_{n \in \mathbb{N}}$ of natural numbers such that $T_X^{\epsilon i_n}(\pi(c_n)) \rightarrow x$. Then the desired result follows from the fact that $|[\epsilon i_n - i, \epsilon i_n + i] \setminus \pi(c_n)^{-1}(\{\infty\})| \leq 1$ for all sufficiently large $n \in \mathbb{N}$. $\square$

Lemma 1.5 ensures that the saturation of $\pi[2^{\mathbb{N}}]$ is co-countable in F , and it is dense by definition. It also ensures that if $F' \subseteq F$ is finite, then $\bigcup \{\mathbb{Z} \setminus x^{-1}(\{\infty\}) \mid x \in F'\}$ is contained in a finite union of translates of I with a finite set. As the defining property of I ensures that such a set has arbitrarily long gaps, it follows that every finite sequence of elements of F is proximal. $\square$

A set $Y \subseteq X$ is a G -clique if all distinct elements of Y are G -related, and G -independent if no elements of Y are G -related.

Proposition 1.6 (ZF + DC). *Suppose that $d \geq 2$, G is an F_σ graph on $2^{\mathbb{N}}$, $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ is a homeomorphism, $P \subseteq 2^{\mathbb{N}}$ is a non-empty perfect*

d-scrambled set, and there is a continuous embedding $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ of $\text{Hyp}^d(G)$ into the *d*-ary Li–Yorkeness relation for which the saturation of $\pi[2^{\mathbb{N}}]$ is co-countable and dense. Then *G* has a perfect clique.

Proof. Fix $i \in \mathbb{Z}$ such that $P \cap (T^i \circ \pi)[2^{\mathbb{N}}]$ is uncountable. Galvin’s Theorem (see, for example, [Kec95, Theorem 19.7]) then yields a perfect set $Q \subseteq (T^i \circ \pi)^{-1}(P)$ that is either a *G*-clique or *G*-independent. But *G*-independence would contradict the fact that *P* is *d*-scrambled. $\square$

2. *D*-SCRAMBLED SETS

We begin this section with the proof of our main result:

Proof (of Theorem 1). By [KV12, Theorem 2.1] and a straightforward Baire category argument, there is an F_σ graph *G* on $2^{\mathbb{N}}$ with an uncountable clique $Y \subseteq 2^{\mathbb{N}}$ but no perfect clique. By Theorem 2, there is a continuous embedding $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ of $\text{Hyp}^d(G)$ into the *d*-ary Li–Yorkeness relation of a homeomorphism $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ for which the saturation of $\pi[2^{\mathbb{N}}]$ is co-countable and dense. Then $\pi[Y]$ is an uncountable *d*-scrambled set and Proposition 1.6 ensures that there is no perfect *d*-scrambled set. $\square$

Let $\mathfrak{c}$ denote the cardinality of the continuum.

Theorem 2.1 (ZFC + $\neg\text{CH}$). *Suppose that $d \geq 3$. Then it is consistent that there is a homeomorphism $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ with a *d*-scrambled set of cardinality $\mathfrak{c}$ but no perfect *d*-scrambled set.*

Proof. By [She99, Theorem 1.13], we can assume that there is an F_σ graph *G* on $2^{\mathbb{N}}$ with a clique of cardinality $\mathfrak{c}$ but no perfect clique. The proof of Theorem 1 therefore yields the desired result. $\square$

A *hypergraph* is a permutation-invariant set of injective sequences.

Theorem 2.2 (ZFC + $\neg\text{CH}$). *Suppose that $d \geq 3$. Then it is consistent that every continuous function on a Polish space with a *d*-scrambled set of cardinality $\aleph_2$ has a *d*-scrambled perfect set.*

Proof. By [KS03, Proposition 3.4], we can assume that every analytic *d*-ary hypergraph with a clique of cardinality $\aleph_2$ has a perfect clique. But the *d*-ary Li–Yorkeness relation is a Borel (and therefore analytic) *d*-ary hypergraph. $\square$

Remark. Given any set $D \subseteq \mathbb{N} \setminus 3$, the underlying ideas can be interleaved to obtain generalizations of our results to sets that are simultaneously *d*-scrambled for all $d \in D$.

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